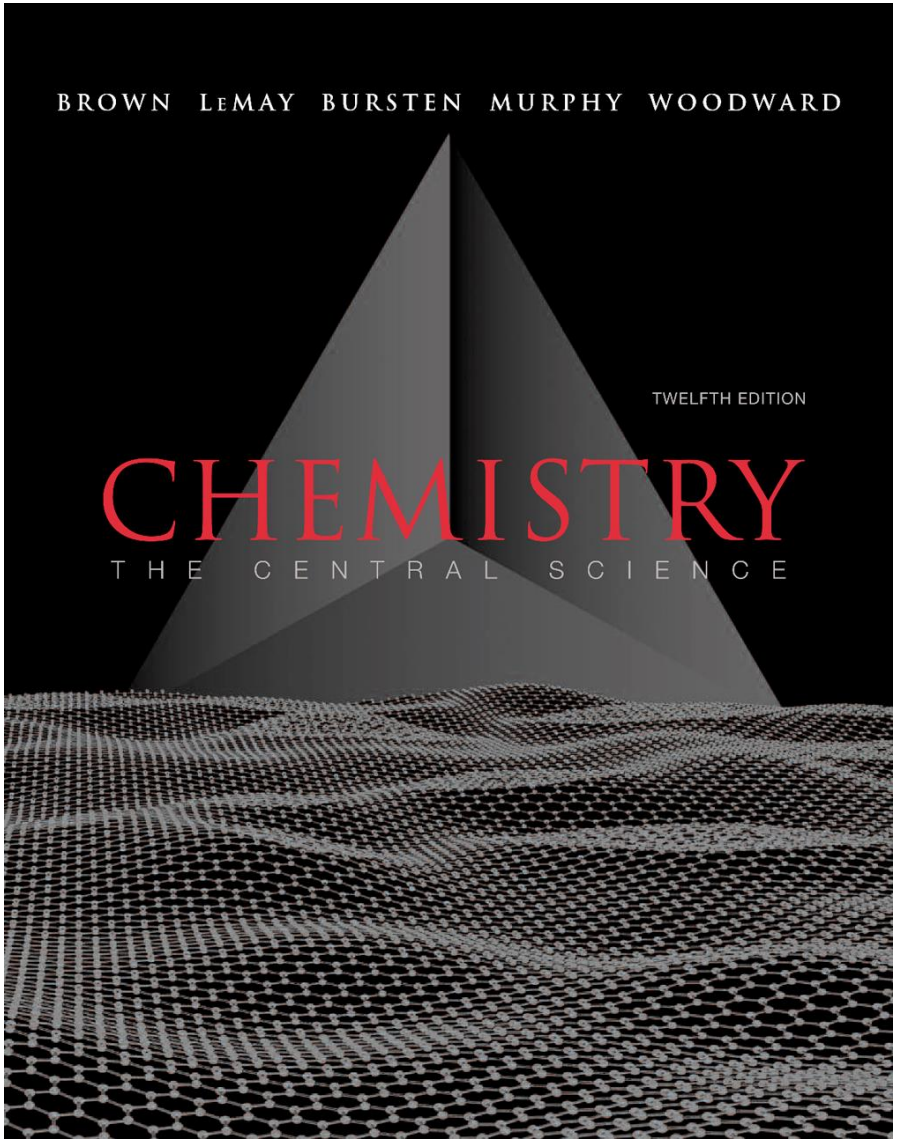


Lecture Presentation

Chapter 2

Atoms, Molecules, and Ions

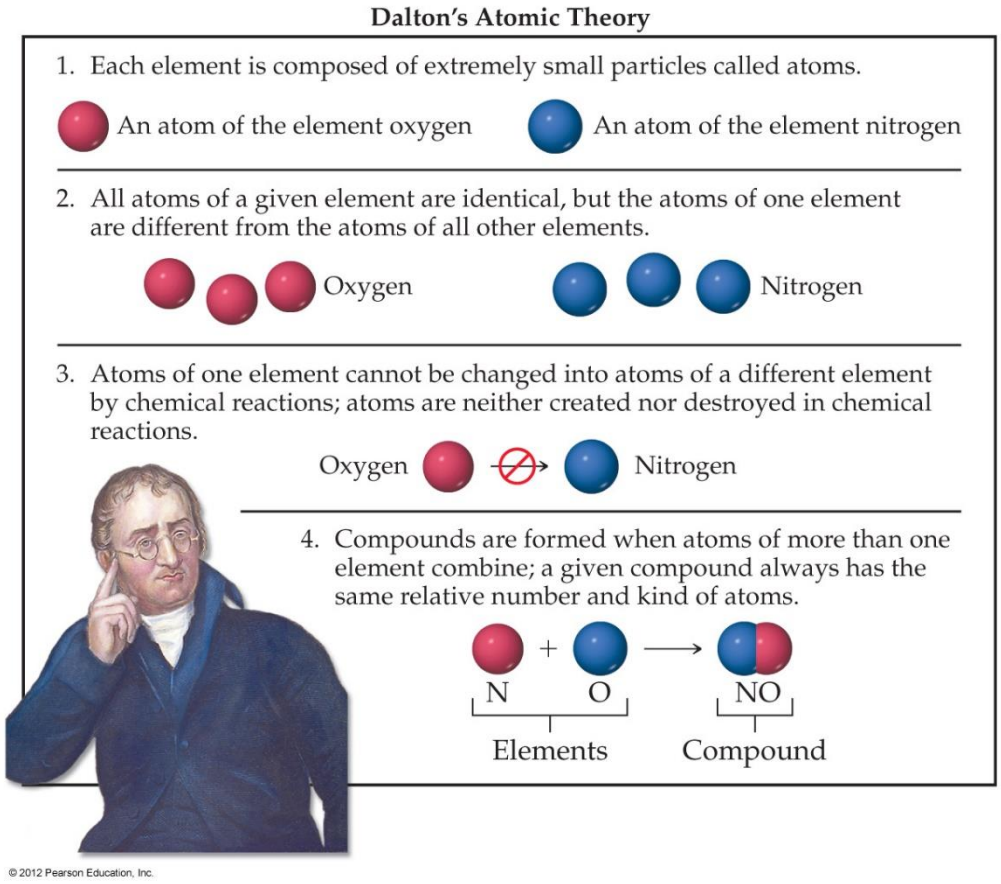
John D. Bookstaver
St. Charles Community College
Cottleville, MO

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Look around at the great variety of colors, textures, and other properties in the materials that surround you—the colors in a garden, the texture of the fabric in your clothes, the solubility of sugar in a cup of coffee, or the beauty and complexity of a geode like the one shown to the right. How can we explain the striking and seemingly infinite variety of properties of the materials that make up our world? What makes diamonds transparent and hard? A large crystal of sodium chloride, table salt, looks a bit like a diamond, but is brittle and readily dissolves in water. What accounts for the differences? Why does paper burn, and why does water quench fires? The answers to all such questions lie in the structures of atoms, which determine the physical and chemical properties of matter.

The notion of **atoms** reemerged in Europe during the seventeenth century. As chemists learned to measure the amounts of elements that reacted with one another to form new substances, the ground was laid for an atomic theory that linked the idea of elements with the idea of atoms. That theory came from the work of John Dalton during the period from 1803 to 1807. Dalton's atomic theory was based on four postulates (see ▼ Figure 2.1).

Atomic Theory of Matter

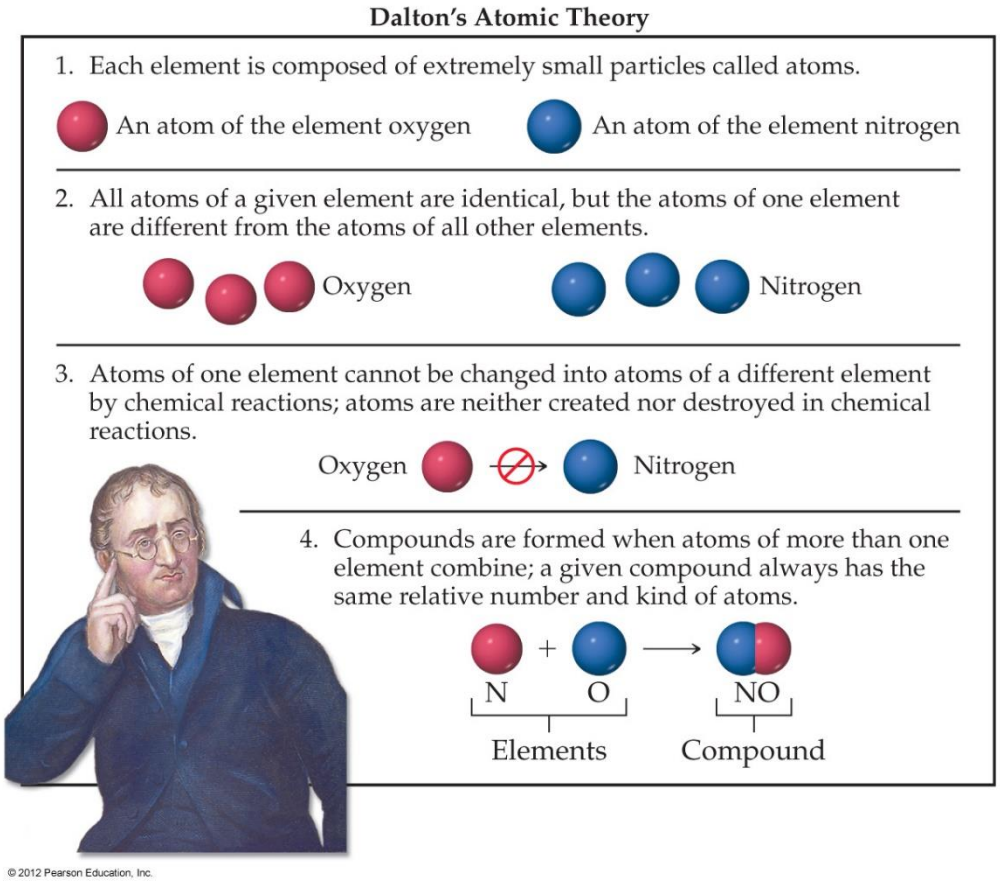


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The theory that atoms are the fundamental building blocks of matter reemerged in the early nineteenth century, championed by John Dalton.

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Dalton's Postulates

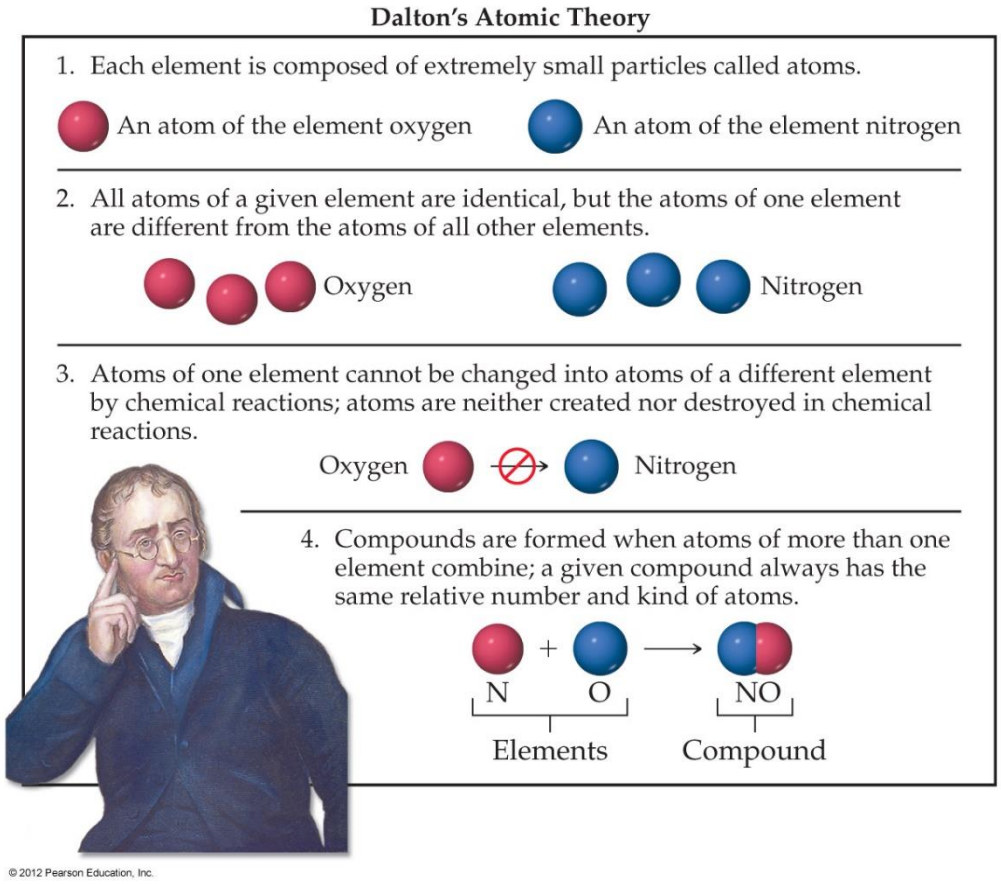


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Each element is composed of extremely small particles called **atoms**.

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Dalton's Postulates

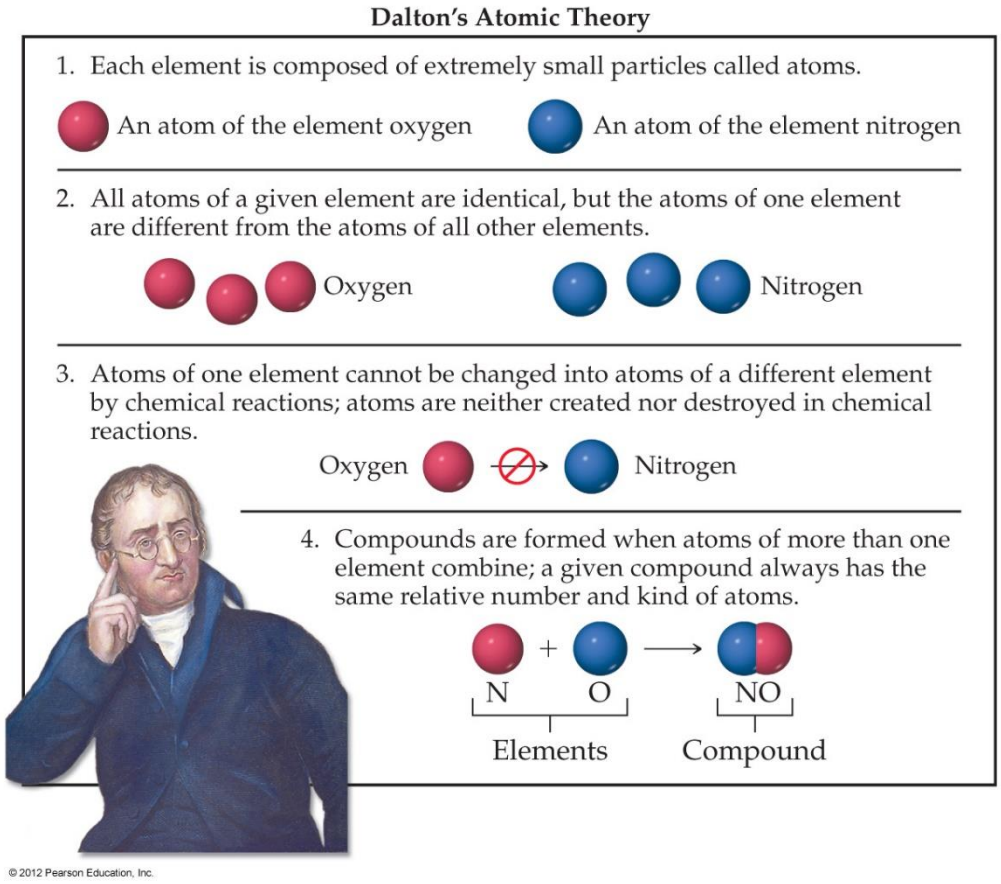


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All atoms of a given element are identical to one another in mass and other properties, but the atoms of one element are different from the atoms of all other elements.

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Dalton's Postulates

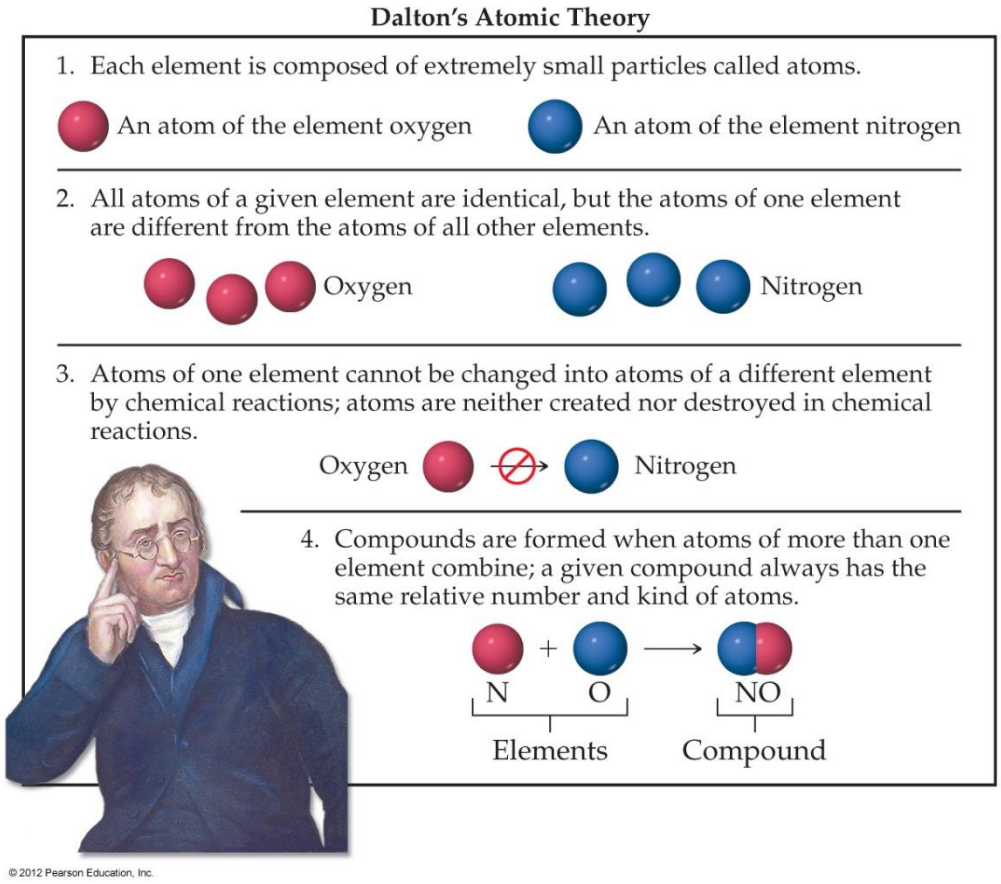


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Atoms of an element are not changed into atoms of a different element by chemical reactions; atoms are neither created nor destroyed in chemical reactions.

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Dalton's Postulates



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Compounds are formed when atoms of more than one element combine; a given compound always has the same relative number and kind of atoms.

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Law of Conservation of Mass

The **total mass** of substances present at the end of a chemical process is the same as the mass of substances present before the process took place.

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law of conservation of mass

The total mass of materials present after a chemical reaction is the same as the total mass present before the reaction.

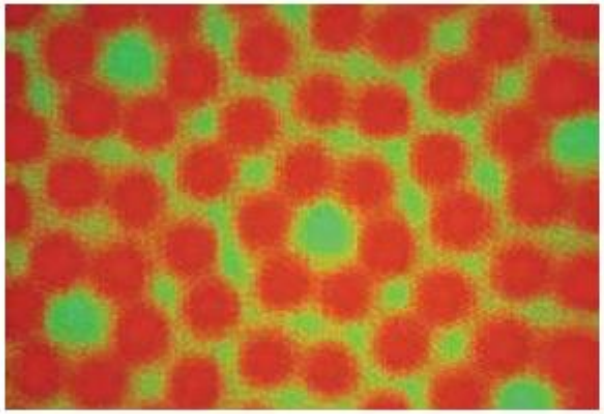
law of constant composition

In a given compound, the relative numbers and kinds of atoms are constant.

law of multiple proportions

If two elements A and B combine to form more than one compound, the masses of B that can combine with a given mass of A are in the ratio of small whole numbers.

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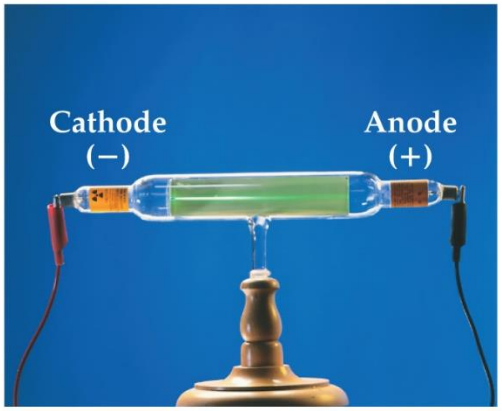
▲ **Figure 2.2 An image of the surface of silicon.** The image was obtained by a technique called scanning tunneling microscopy. The color was added to the image by computer to help distinguish its features. Each red sphere is a silicon atom.

As scientists developed methods for probing the nature of matter, the supposedly indivisible atom began to show signs of a more complex structure, and today we know that the atom is composed of subatomic particles.

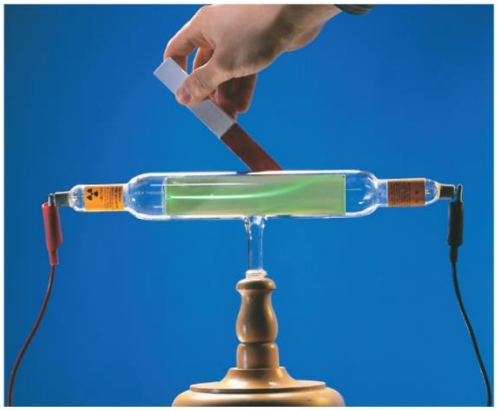
atom is composed in part of electrically charged particles, some with a positive charge and some with a negative charge.

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The Electron



(a) Electrons move from the cathode (negative electrode) to the anode (positive electrode). The tube contains a glass screen (set diagonally to the electron beam) that fluoresces, showing the path of the cathode rays.



(b) The rays are deflected by a magnet.

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- Streams of negatively charged particles were found to emanate from cathode tubes, causing fluorescence.
- J. J. Thomson is credited with their discovery (1897).

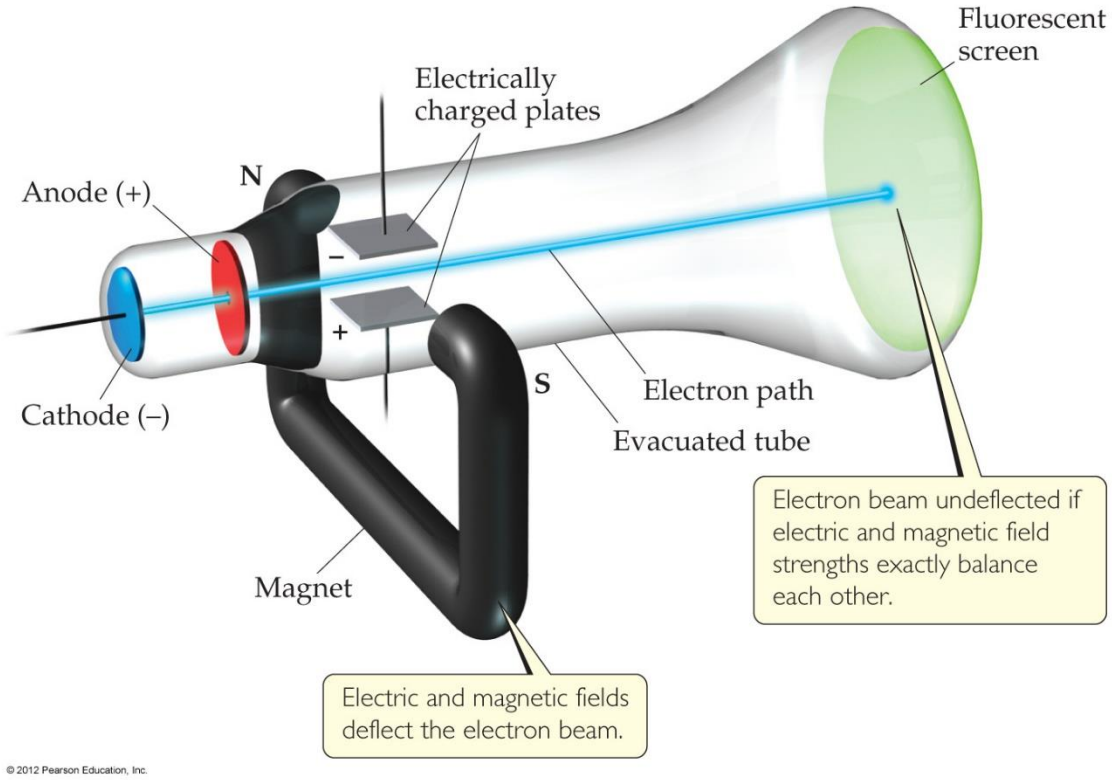
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Thomson constructed a cathode-ray tube having a hole in the anode through which a beam of electrons passed. Electrically charged plates and a magnet were positioned perpendicular to the electron beam, and a fluorescent screen was located at one end.

The electric field deflected the rays in one direction, and the magnetic field deflected them in the opposite direction. Thomson adjusted the strengths of the fields so that the effects balanced each other, allowing the electrons to travel in a straight path to the screen.

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The Electron

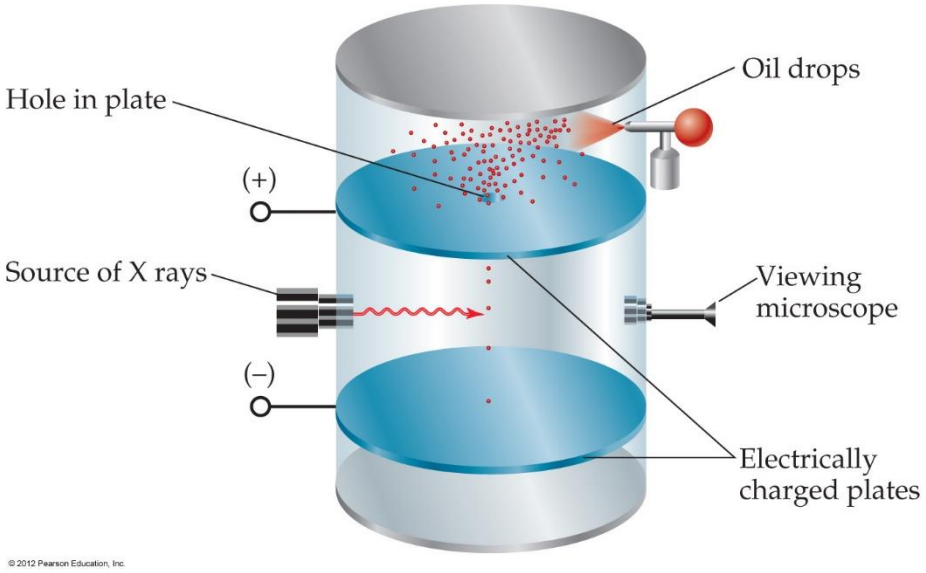


Thomson measured the charge/mass ratio of the electron to be 1.76×10^8 coulombs/gram (C/g).

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Millikan Oil-Drop Experiment

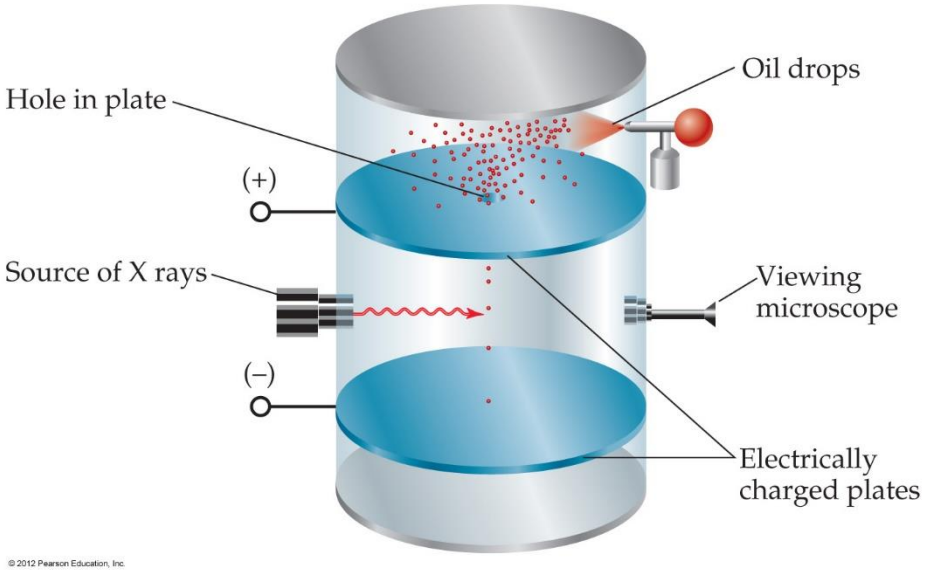
Once the charge/mass ratio of the electron was known, determination of either the charge or the mass of an electron would yield the other.



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Millikan Oil-Drop Experiment

Robert Millikan (University of Chicago) determined the charge on the electron in 1909.



Because the charge on any drop was always some integral multiple of 1.602×10^{-19} C, Millikan deduced this value to be the charge of a single electron.

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Radioactivity

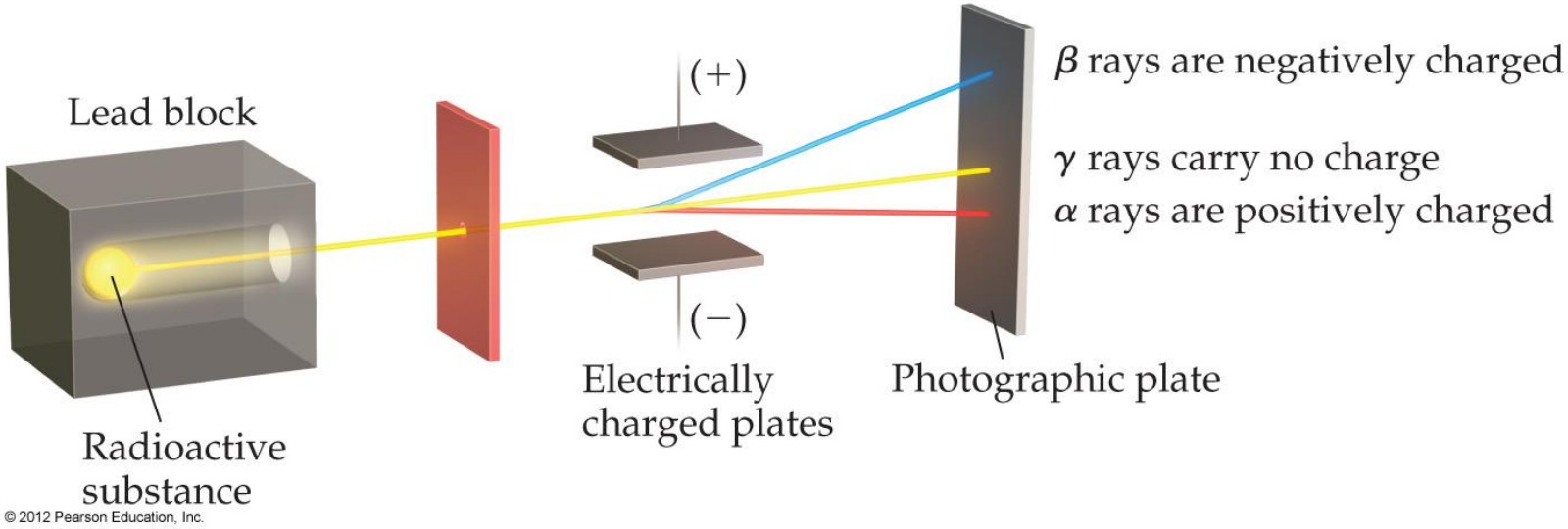
In 1896 the French scientist Henri Becquerel (1852–1908) discovered that a compound of uranium spontaneously emits high-energy radiation.

- **Radioactivity** is the spontaneous emission of radiation by an atom.
- Marie and Pierre Curie also studied it.

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Radioactivity

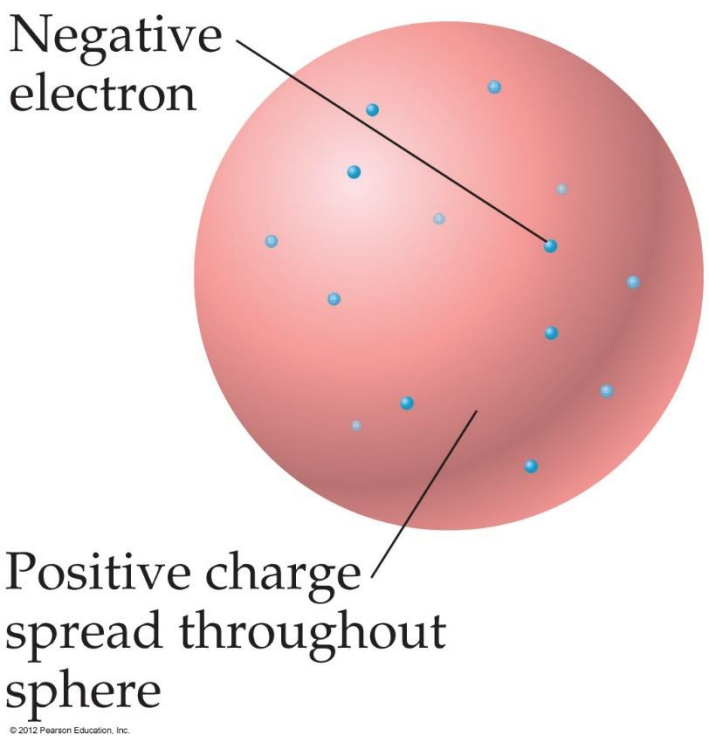
- Three types of radiation were discovered by Ernest Rutherford:
  - α particles
  - β particles
  - γ rays



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The Atom

Thomson proposed that the atom consists of a uniform positive sphere of matter in which the mass is evenly distributed and in which the electrons are embedded like raisins in a pudding or seeds in a watermelon.

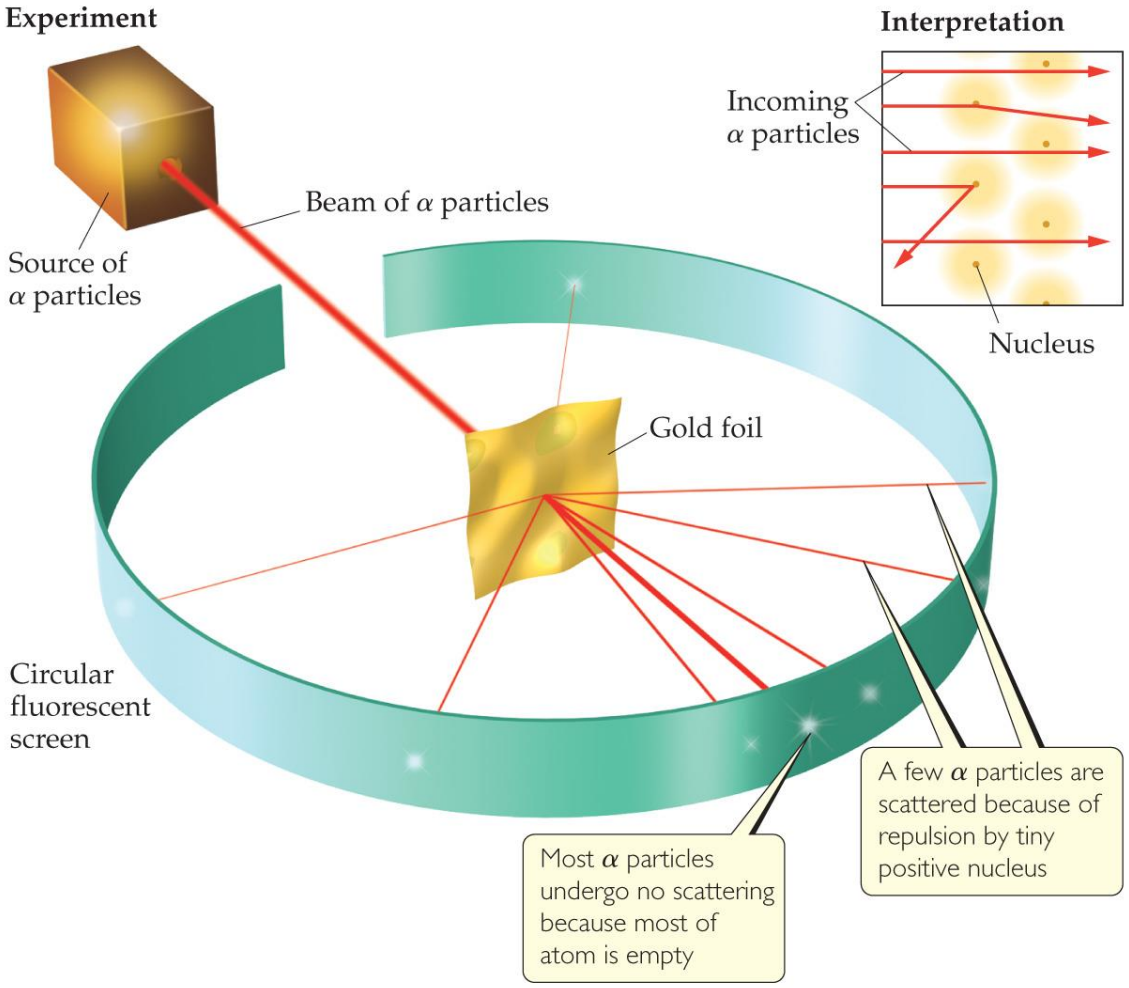


- The prevailing theory was that of the “plum pudding” model, put forward by Thomson.
- It featured a positive sphere of matter with negative electrons imbedded in it.

Ernest Rutherford and Ernest Marsden proved this model wrong.

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Discovery of the Nucleus

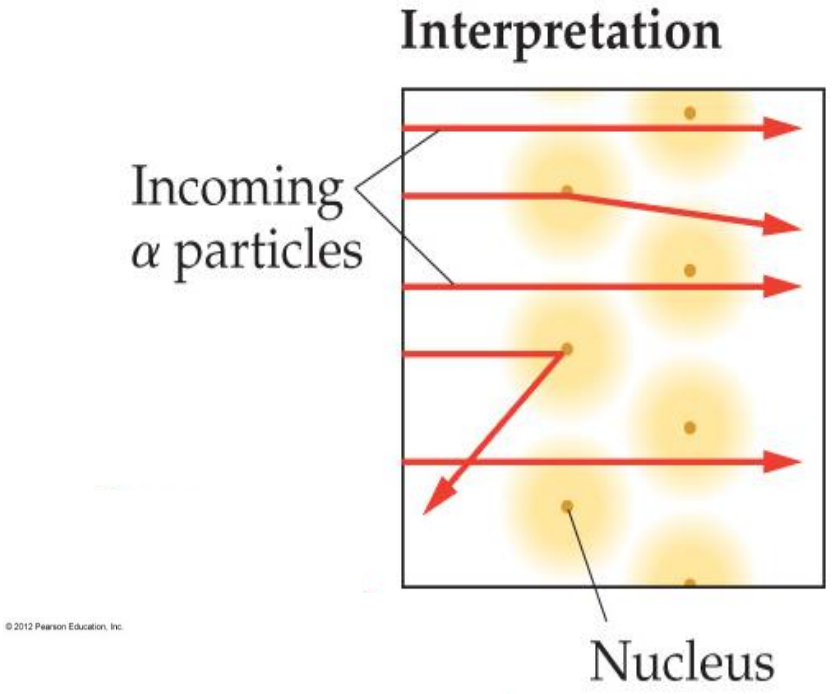


Ernest Rutherford shot α particles at a thin sheet of gold foil and observed the pattern of scatter of the particles.

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The Nuclear Atom

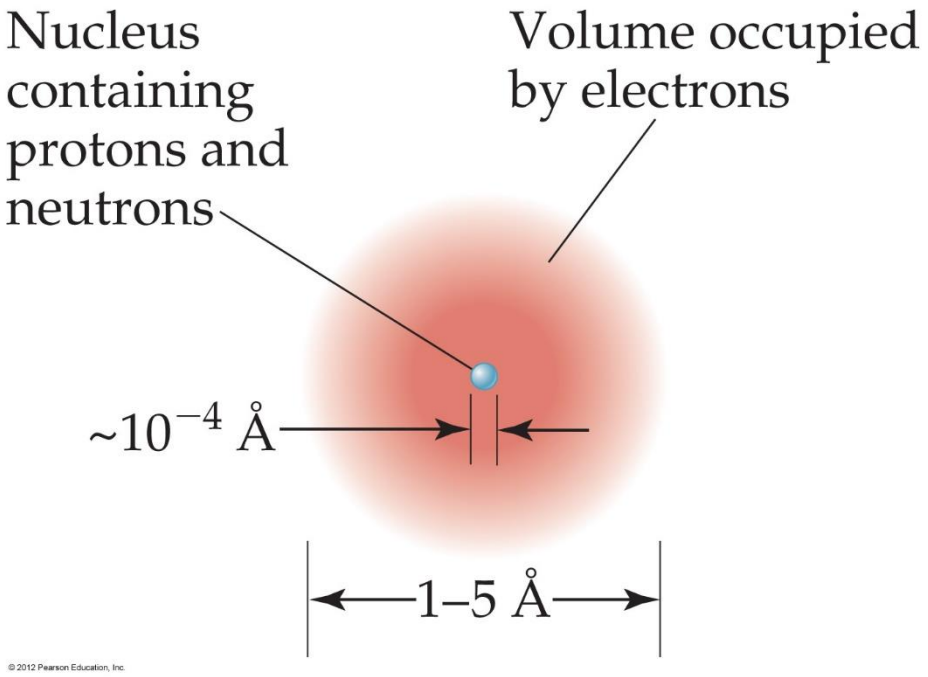
When a particles pass through a gold foil, most pass through undeflected but some are scattered, a few at very large angles. According to the plum-pudding model of the atom, the particles should experience only very minör deflections. The nuclear model of the atom explains why a few a particles are deflected at large angles. although the nuclear atom has been depicted here as a yellow sphere, it is important to
realize that most of the space around the nucleus contains only the low mass electrons.



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The Nuclear Atom

- Rutherford postulated a very small, dense nucleus with the electrons around the outside of the atom.
- Most of the volume of the atom is empty space.



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Other Subatomic Particles

- Protons were discovered by Rutherford in 1919.
- Neutrons were discovered by James Chadwick in 1932.

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Subatomic Particles

- Protons and electrons are the only particles that have a charge.
- Protons and neutrons have essentially the same mass.
- The mass of an electron is so small we ignore it.

TABLE 2.1 • Comparison of the Proton, Neutron, and Electron

Particle	Charge	Mass (amu)
Proton	Positive (1+)	1.0073
Neutron	None (neutral)	1.0087
Electron	Negative (1−)	5.486×10^{-4}

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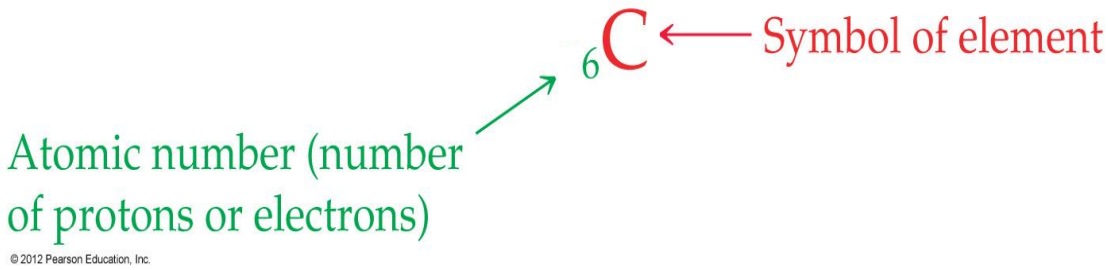
Symbols of Elements



Elements are symbolized by one or two letters.

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Symbols of Elements

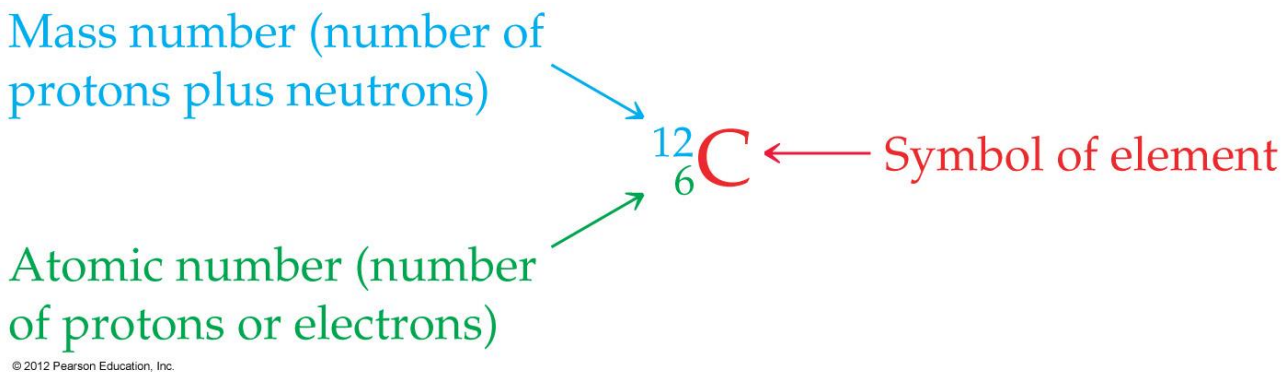


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All atoms of the same element have the same number of protons, which is called the **atomic number**, *Z*.

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Symbols of Elements



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The mass of an atom in atomic mass units (amu) is the total number of protons and neutrons in the atom.

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Isotopes

- **Isotopes** are atoms of the same element with different masses.
- Isotopes have different numbers of neutrons.

TABLE 2.2 • Some Isotopes of Carbon*

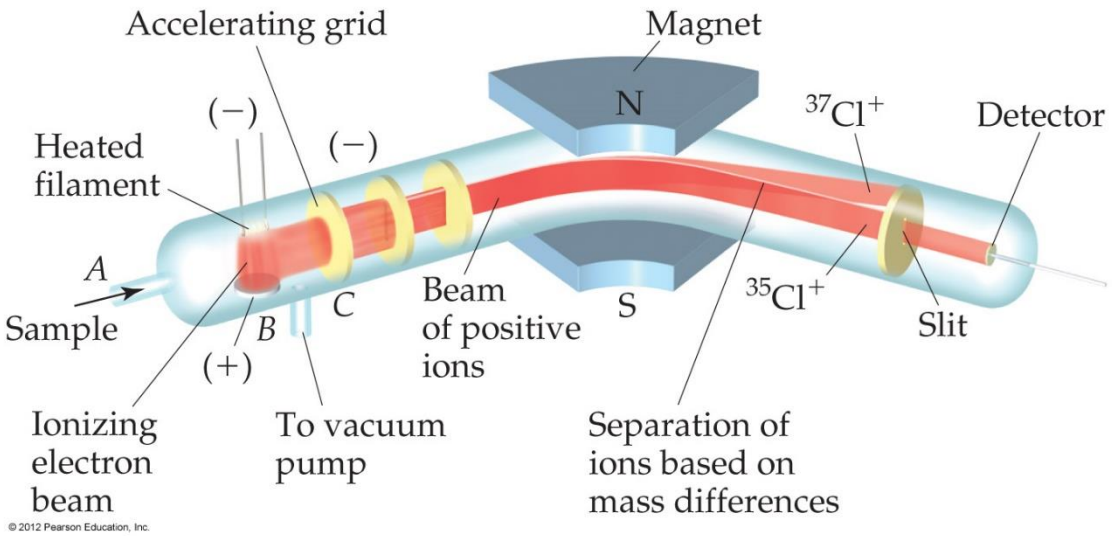
Symbol	Number of Protons	Number of Electrons	Number of Neutrons
^{11}C	6	6	5
^{12}C	6	6	6
^{13}C	6	6	7
^{14}C	6	6	8

*Almost 99% of the carbon found in nature is ^{12}C.

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Atomic Mass



Atomic and molecular masses can be measured with great accuracy using a mass spectrometer.

Today we can determine the masses of individual atoms with a high degree of accuracy. For example, we know that the ^1H atom has a mass of 1.6735×10^{-24} g and the ^{16}O atom has a mass of 2.6560×10^{-23} g. As we noted in Section 2.3, it is convenient to use the **atomic mass unit** when dealing with these extremely small masses:

$$1 \text{ amu} = 1.66054 \times 10^{-24} \text{ g and } 1 \text{ g} = 6.02214 \times 10^{23} \text{ amu}$$

The atomic mass unit is presently defined by assigning a mass of exactly 12 amu to a chemically unbound atom of the ^{12}C isotope of carbon. In these units, an ^1H atom has a mass of 1.0078 amu and an ^{16}O atom has a mass of 15.9949 amu.

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Average Mass

- Because in the real world we use large amounts of atoms and molecules, we use average masses in calculations.
- **Average mass** is calculated from the isotopes of an element weighted by their relative abundances.

Atomic Weight

Most elements occur in nature as mixtures of isotopes. We can determine the *average atomic mass* of an element, usually called the element's **atomic weight**, by summing (indicated by the Greek sigma, $\sum$) over the masses of its isotopes multiplied by their relative abundances:

$$\text{Atomic weight} = \sum_{\text{over all isotopes of the element}} [(\text{isotope mass}) \times (\text{fractional isotope abundance})] \quad [2.1]$$

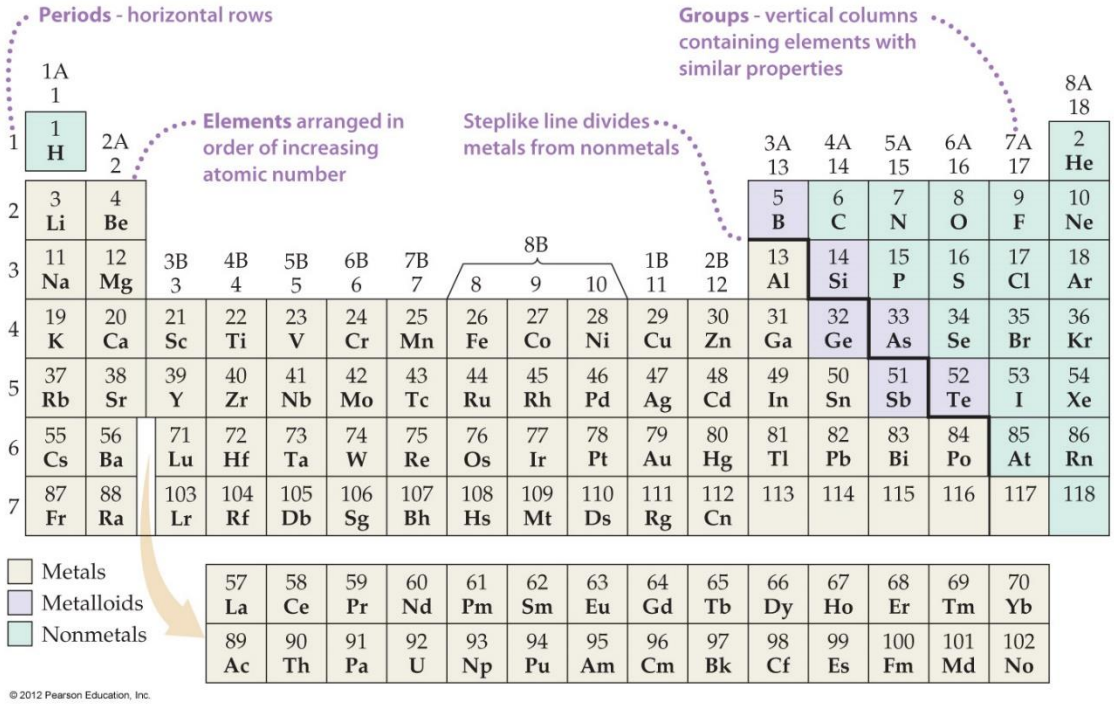
Naturally occurring carbon, for example, is composed of 98.93% ^{12}C and 1.07% ^{13}C. The masses of these isotopes are 12 amu (exactly) and 13.00335 amu, respectively, making the atomic weight of carbon

$$(0.9893)(12\ \text{amu}) + (0.0107)(13.00335\ \text{amu}) = 12.01\ \text{amu}$$

The atomic weights of the elements are listed in both the periodic table and the table of elements front inside cover of this text.

s,
les,
ns

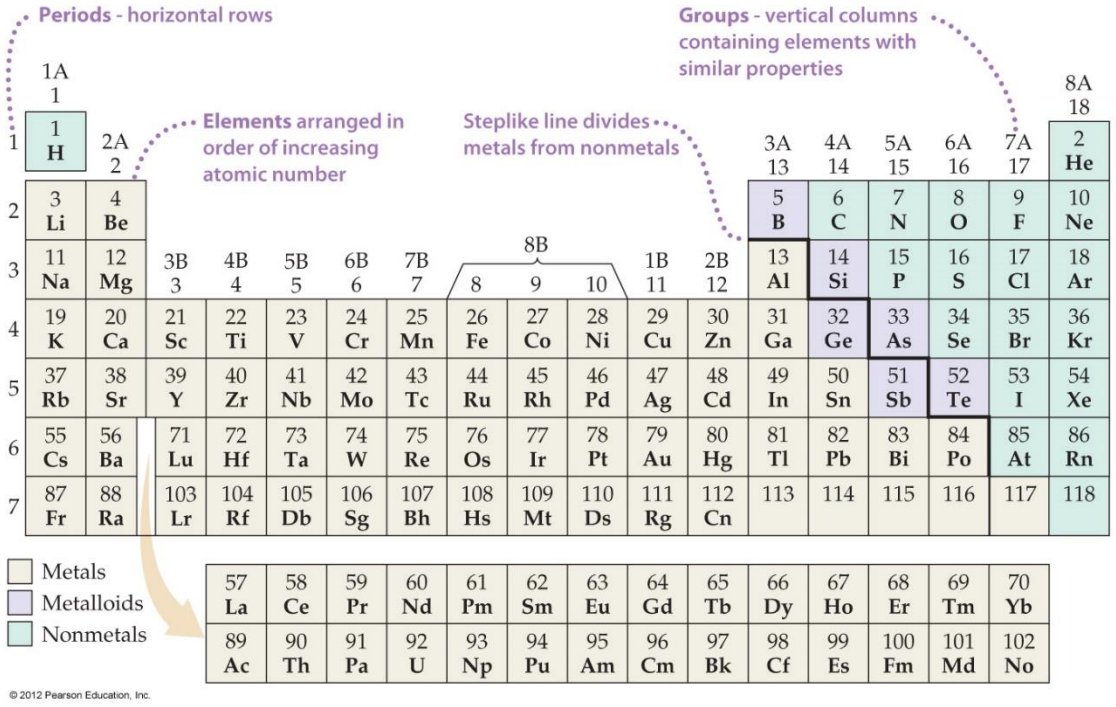
Periodic Table



- The **periodic table** is a systematic catalog of the elements.
- Elements are arranged in order of atomic number.

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Periodic Table

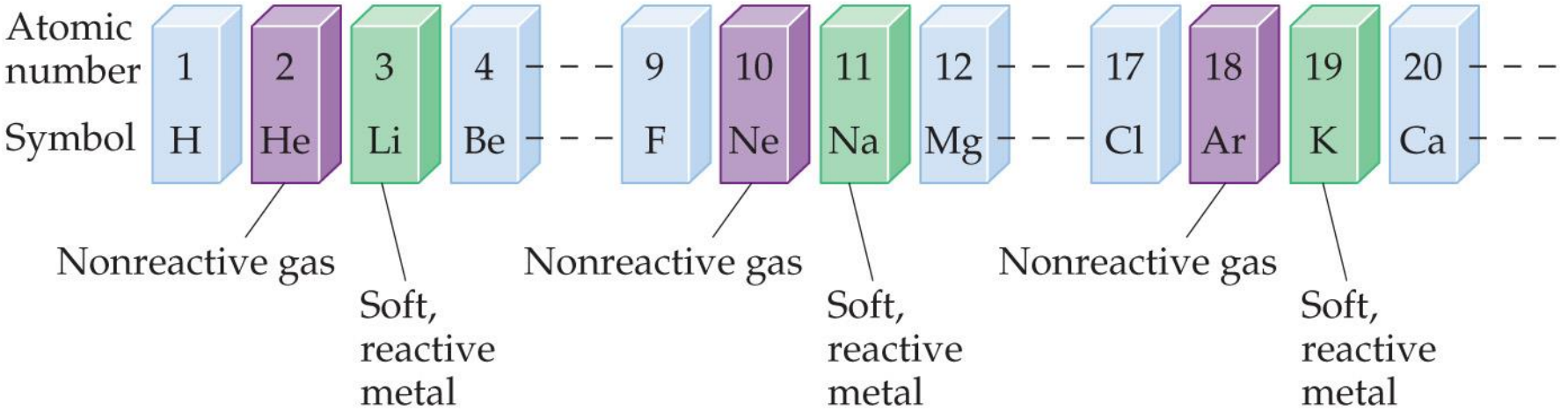


- The rows on the periodic chart are periods.
- Columns are groups.
- Elements in the same group have similar chemical properties.

Atoms, Molecules, and Ions

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Periodicity



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When one looks at the chemical properties of elements, one notices a repeating pattern of reactivities.

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Groups

TABLE 2.3 • Names of Some Groups in the Periodic Table

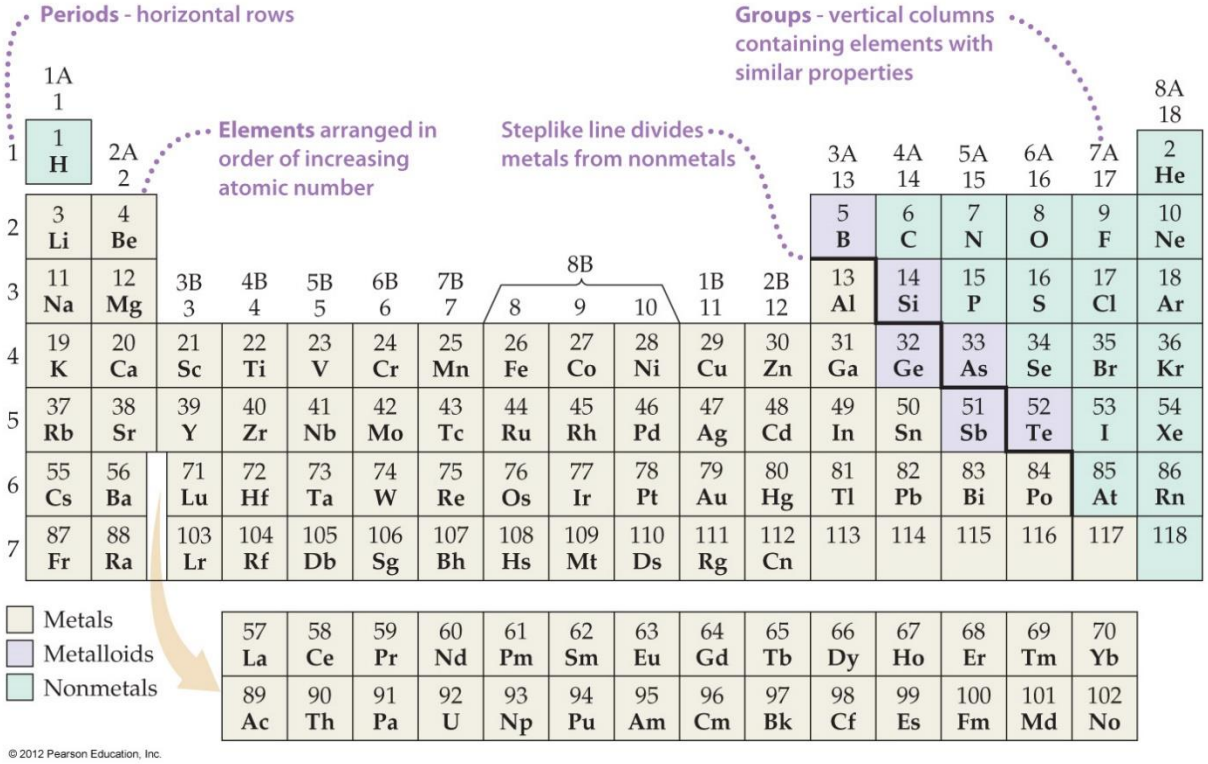
Group	Name	Elements
1A	Alkali metals	Li, Na, K, Rb, Cs, Fr
2A	Alkaline earth metals	Be, Mg, Ca, Sr, Ba, Ra
6A	Chalcogens	O, S, Se, Te, Po
7A	Halogens	F, Cl, Br, I, At
8A	Noble gases (or rare gases)	He, Ne, Ar, Kr, Xe, Rn

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These five groups are known by their names.

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Periodic Table



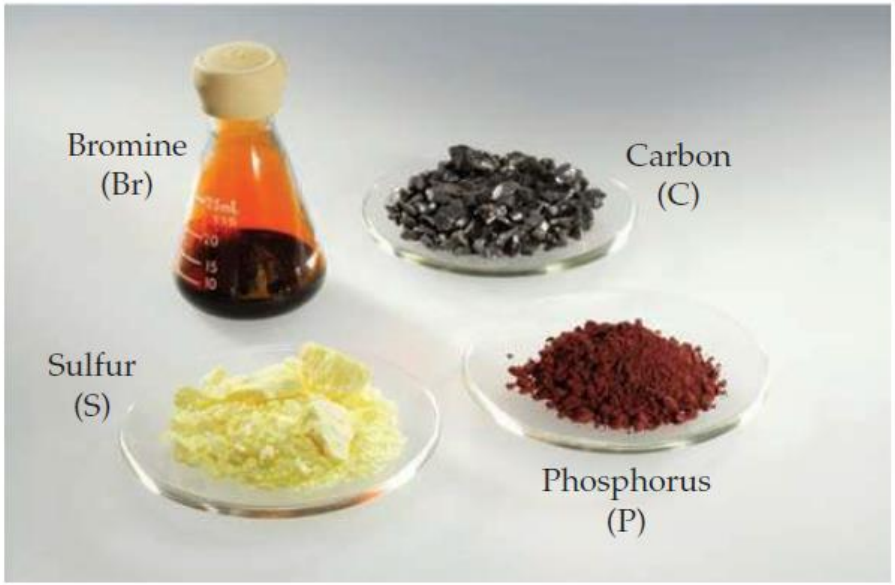
Nonmetals are on the right side of the periodic table (with the exception of H).

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Metals

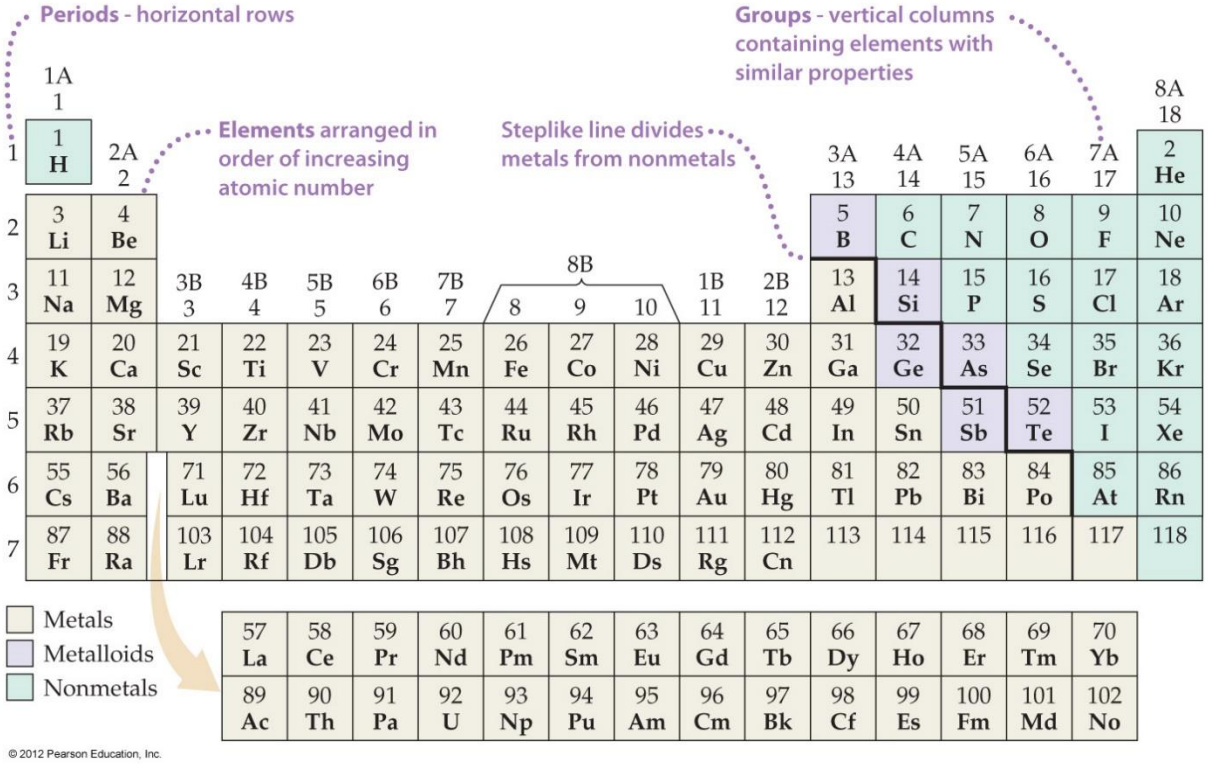


Nonmetals



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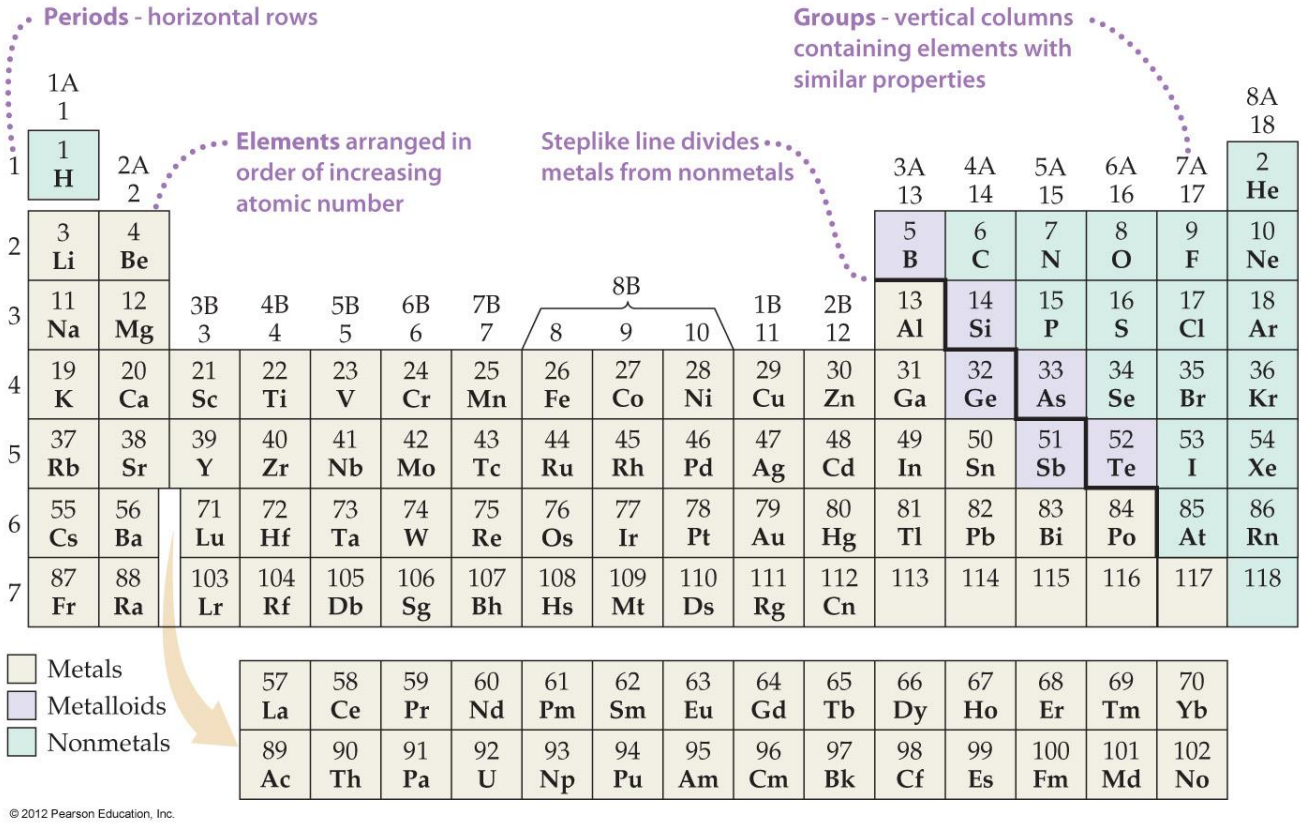
Periodic Table



Metalloids border the stair-step line (with the exception of Al, Po, and At).

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Periodic Table



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Metals are on the left side of the chart.

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Chemical Formulas



Hydrogen, H_2



Oxygen, O_2



Water, H_2O



Hydrogen peroxide, H_2O_2



Carbon monoxide, CO



Carbon dioxide, CO_2



Methane, CH_4



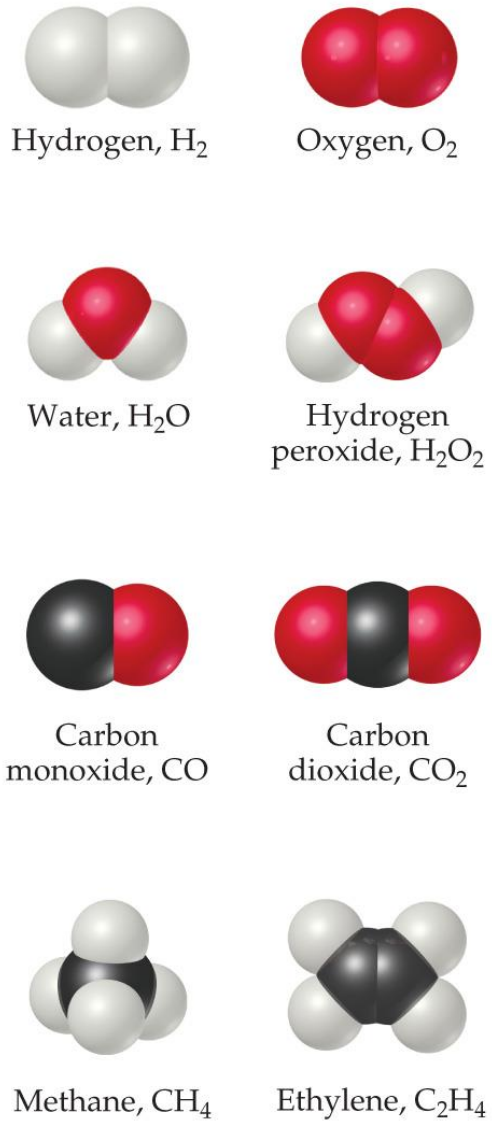
Ethylene, C_2H_4

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The subscript to the right of the symbol of an element tells the number of atoms of that element in one molecule of the compound.

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Chemical Formulas



Molecular compounds are composed of molecules and almost always contain only nonmetals.

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Diatomic Molecules

- These seven elements occur naturally as molecules containing two atoms:
  - Hydrogen
  - Nitrogen
  - Oxygen
  - Fluorine
  - Chlorine
  - Bromine
  - Iodine

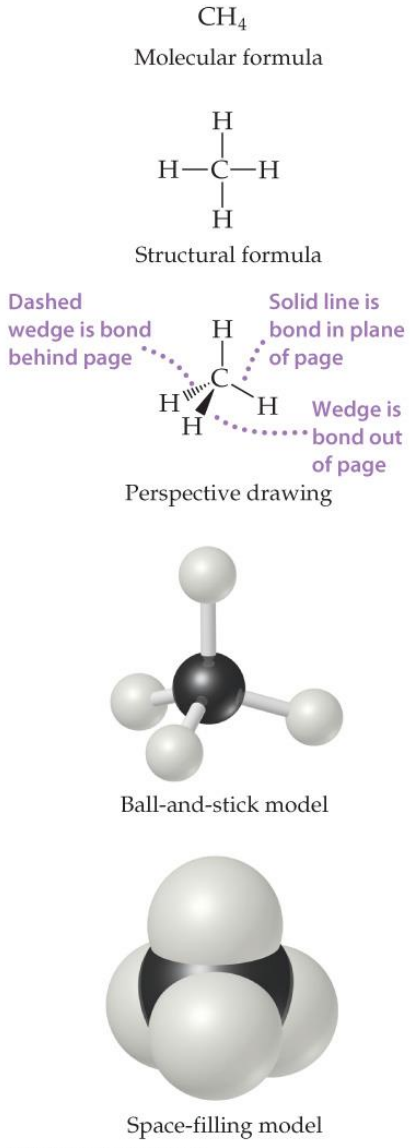
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Types of Formulas

- **Empirical formulas** give the lowest whole-number ratio of atoms of each element in a compound.
- **Molecular formulas** give the exact number of atoms of each element in a compound.

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Types of Formulas

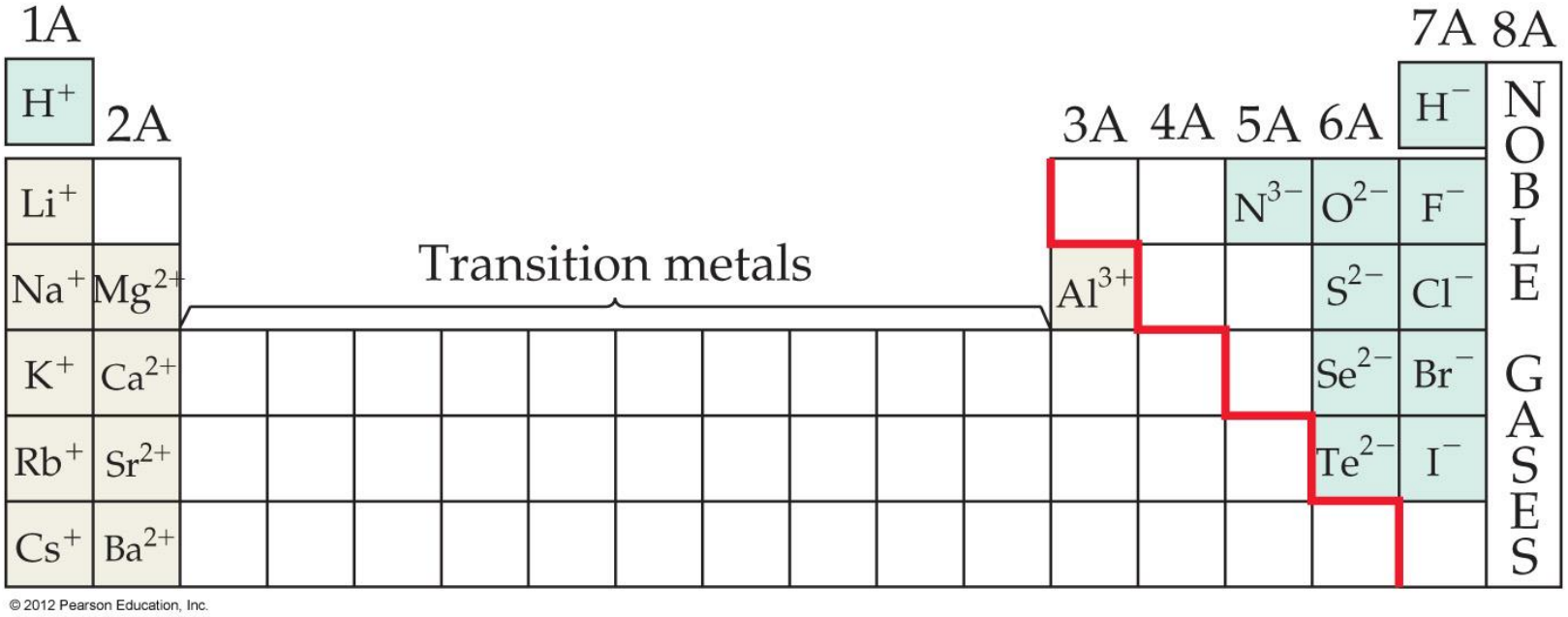


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- **Structural formulas** show the order in which atoms are bonded.
- **Perspective drawings** also show the three-dimensional array of atoms in a compound.

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Ions



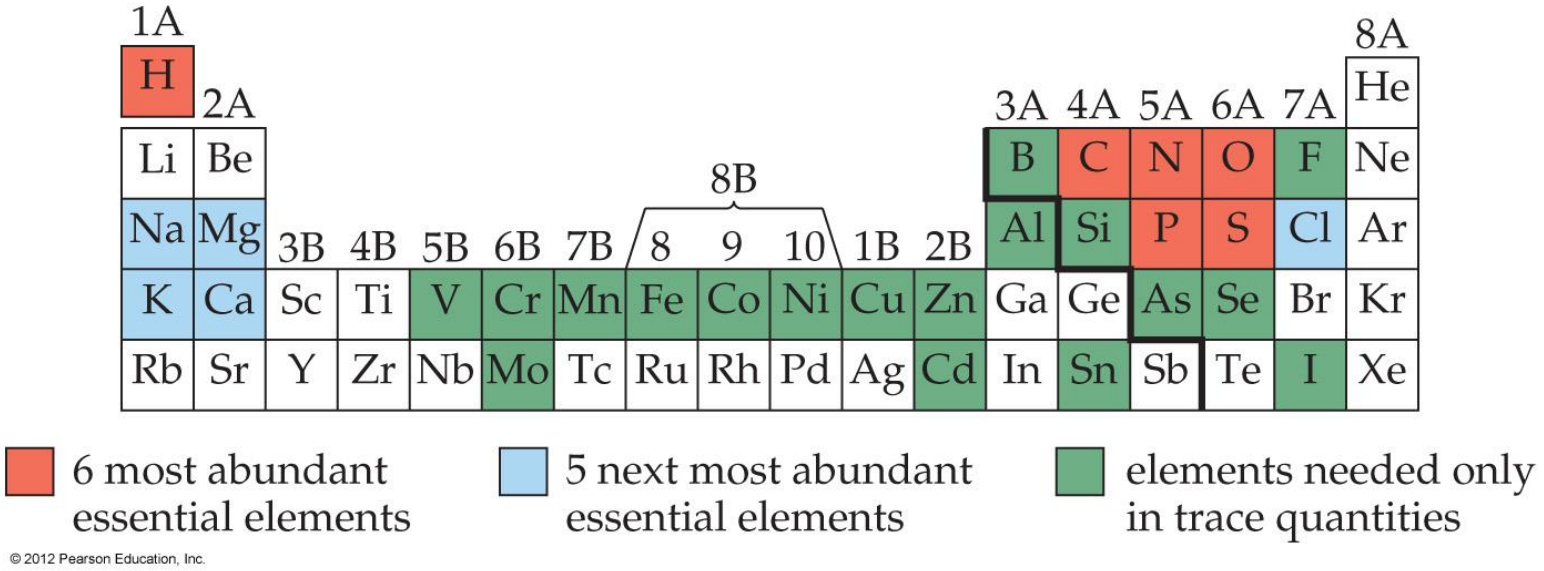
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- When atoms lose or gain electrons, they become **ions**.
  - **Cations** are positive and are formed by elements on the left side of the periodic chart.
  - **Anions** are negative and are formed by elements on the right side of the periodic chart.

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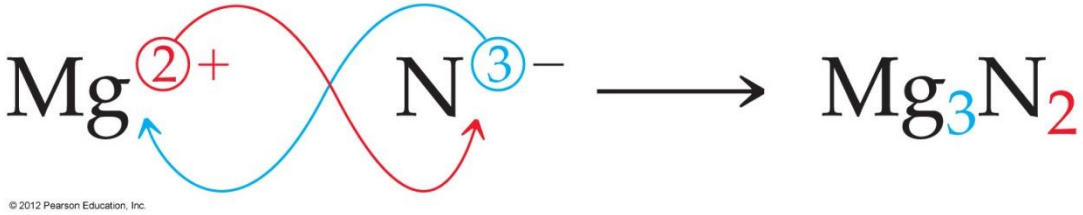
Ionic Bonds

Ionic compounds (such as NaCl) are generally formed between metals and nonmetals.



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Writing Formulas



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- Because compounds are electrically neutral, one can determine the formula of a compound this way:
  - The charge on the cation becomes the subscript on the anion.
  - The charge on the anion becomes the subscript on the cation.
  - If these subscripts are not in the lowest whole-number ratio, divide them by the greatest common factor.

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Common Cations

TABLE 2.4 • Common Cations*

Charge	Formula	Name	Formula	Name
1+	**H^+**	**hydrogen ion**	**NH_4^+**	**ammonium ion**
	Li^+	lithium ion	Cu^+	copper(I) or cuprous ion
	Na^+	**sodium ion**		
	K^+	**potassium ion**		
	Cs^+	cesium ion		
	Ag^+	**silver ion**		
2+	**Mg^{2+}**	**magnesium ion**	Co^{2+}	cobalt(II) or cobaltous ion
	Ca^{2+}	**calcium ion**	**Cu^{2+}**	**copper(II)** or cupric ion
	Sr^{2+}	strontium ion	**Fe^{2+}**	**iron(II)** or ferrous ion
	Ba^{2+}	barium ion	Mn^{2+}	manganese(II) or manganous ion
	Zn^{2+}	**zinc ion**	Hg_2^{2+}	mercury(I) or mercurous ion
	Cd^{2+}	cadmium ion	**Hg^{2+}**	**mercury(II)** or mercuric ion
			Ni^{2+}	nickel(II) or nickelous ion
			Pb^{2+}	**lead(II)** or plumbous ion
			Sn^{2+}	tin(II) or stannous ion
3+	**Al^{3+}**	**aluminum ion**	Cr^{3+}	chromium(III) or chromic ion
			Fe^{3+}	**iron(III)** or ferric ion

*The ions we use most often in this course are in boldface. Learn them first.

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Common Anions

TABLE 2.5 • Common Anions*

Charge	Formula	Name	Formula	Name
1−	H^-	hydride ion	**CH_3COO^-** (or $C_2H_3O_2^-$)	**acetate ion**
	F^-	**fluoride ion**	ClO_3^-	chlorate ion
	Cl^-	**chloride ion**	**ClO_4^-**	**perchlorate ion**
	Br^-	**bromide ion**	**NO_3^-**	**nitrate ion**
	I^-	**iodide ion**	MnO_4^-	permanganate ion
	CN^-	cyanide ion		
	OH^-	**hydroxide ion**		
2−	**O^{2-}**	**oxide ion**	**CO_3^{2-}**	**carbonate ion**
	O_2^{2-}	peroxide ion	CrO_4^{2-}	chromate ion
	S^{2-}	**sulfide ion**	$Cr_2O_7^{2-}$	dichromate ion
			SO_4^{2-}	**sulfate ion**
3−	N^{3-}	nitride ion	**PO_4^{3-}**	**phosphate ion**

*The ions we use most often are in boldface. Learn them first.

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Inorganic Nomenclature

Nomen: name
Calare: to call

- Write the name of the cation.
- If the anion is an element, change its ending to ***-ide***; if the anion is a polyatomic ion, simply write the name of the polyatomic ion.
- If the cation can have more than one possible charge, write the charge as a Roman numeral in parentheses.

NaCl: sodium chloride

MgI_2: magnesium iodide

$MgSO_4$: magnesium sulfate

$FeCl_2$: iron(II) chloride

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- For binary non-metal compounds (similar)

HCl: hydrogen chloride
SO_2: sulphur dioxide
SO_3: sulphur trioxide
B_2Br_4: dibor tetrabromide
NO_2: nitrogen dioxide
N_2O_3: dinitrogen trioxide

- For binary acids: hydro…..ic acid

HCl: hydrochloric acid
HBr: hydrobromic acid

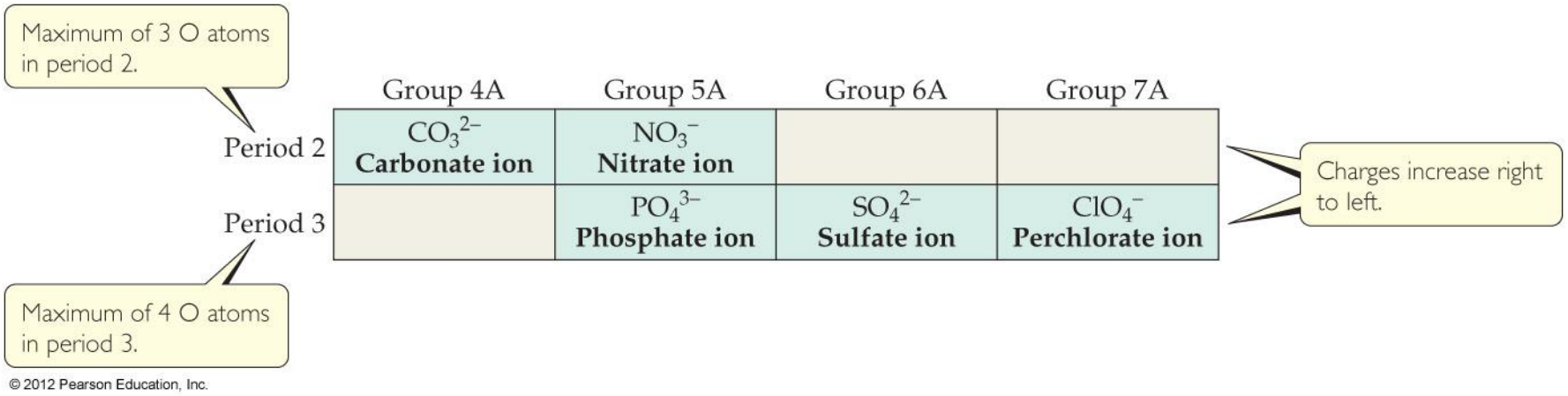
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Patterns in Oxyanion Nomenclature

- When there are two oxyanions involving the same element:
  - The one with fewer oxygens ends in ***-ite***.
  - The one with more oxygens ends in ***-ate***.
    - NO_2^- : nitrite; SO_3^{2-} : sulfite
    - NO_3^- : nitrate; SO_4^{2-} : sulfate

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Patterns in Oxyanion Nomenclature



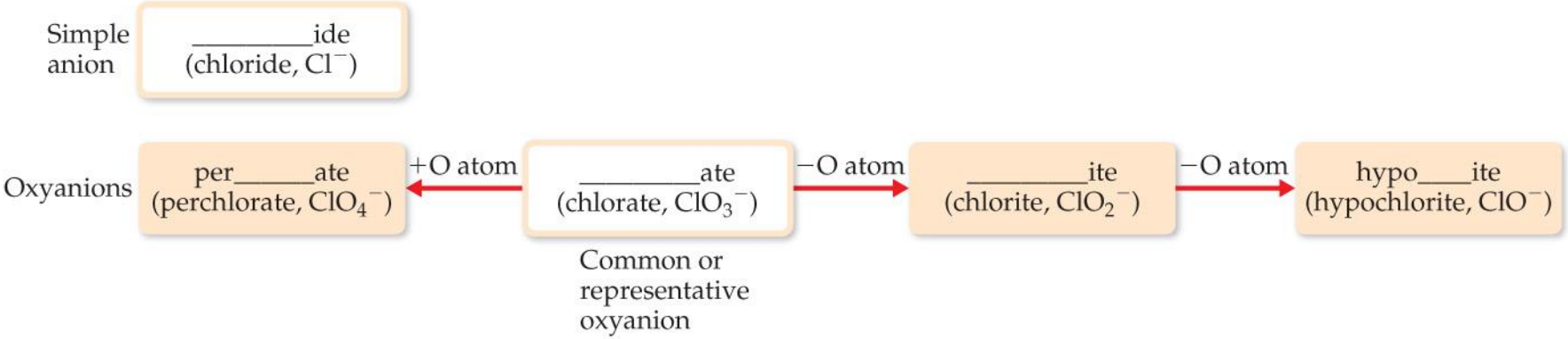
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- Central atoms on the second row have bond to at most three oxygens; those on the third row take up to four.
- Charges increase as you go from *right* to *left*.

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Patterns in Oxyanion Nomenclature

- The one with the second fewest oxygens ends in ***-ite***.
  - ClO_2^- : chlorite
- The one with the second most oxygens ends in ***-ate***.
  - ClO_3^- : chlorate

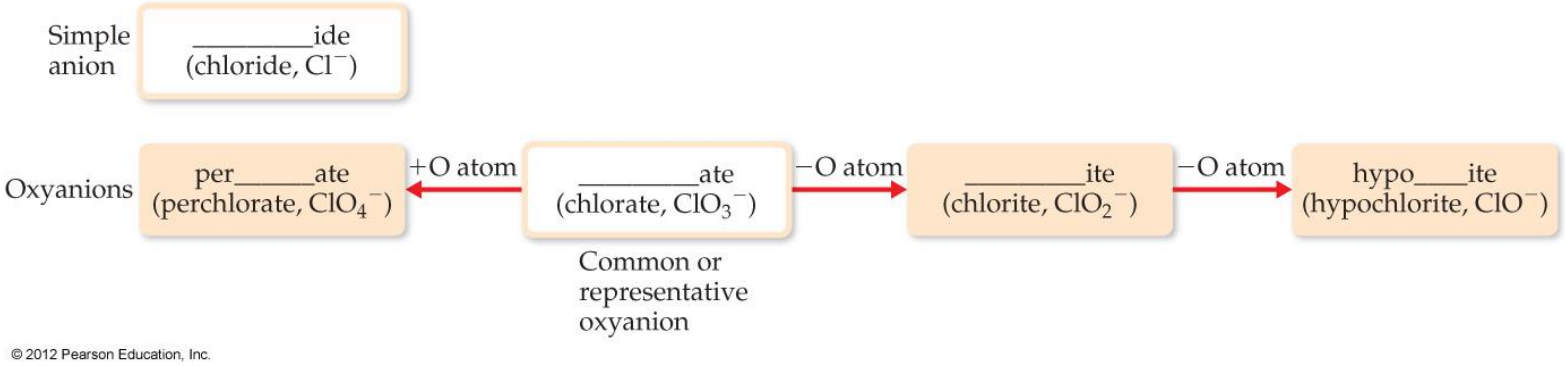


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Patterns in Oxyanion Nomenclature

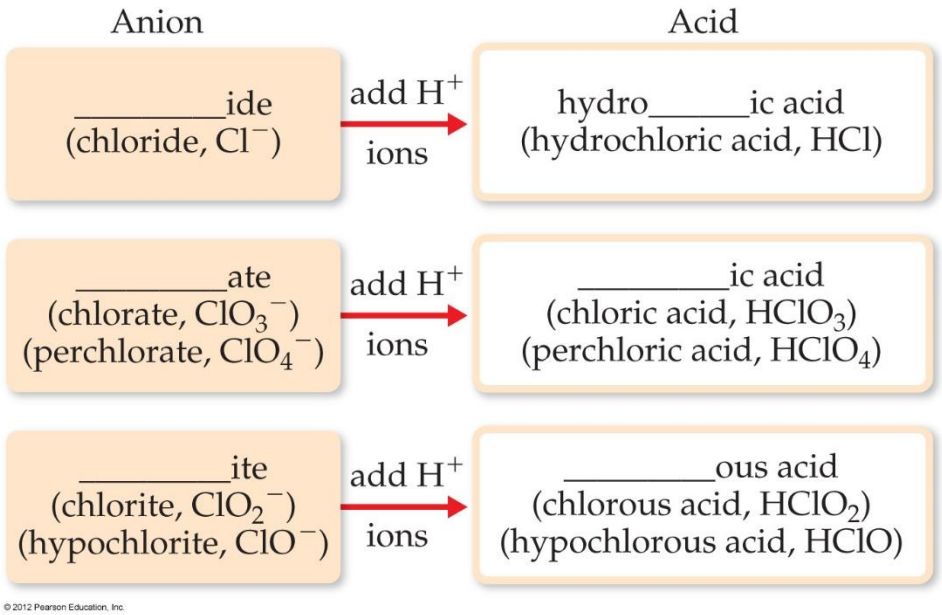
- The one with the fewest oxygens has the prefix ***hypo-*** and ends in ***-ite***.
  - ClO^- : hypochlorite
- The one with the most oxygens has the prefix ***per-*** and ends in ***-ate***.
  - ClO_4^- : perchlorate



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Acid Nomenclature

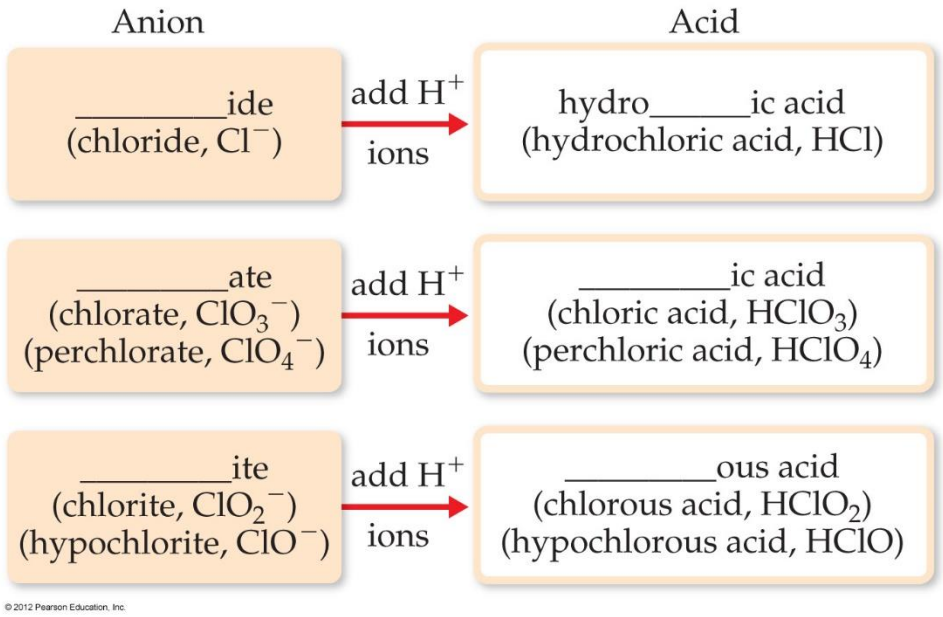


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- If the anion in the acid ends in ***-ide***, change the ending to ***-ic acid*** and add the prefix ***hydro-*** .
  - HCl: hydrochloric acid
  - HBr: hydrobromic acid
  - HI: hydroiodic acid

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Acid Nomenclature

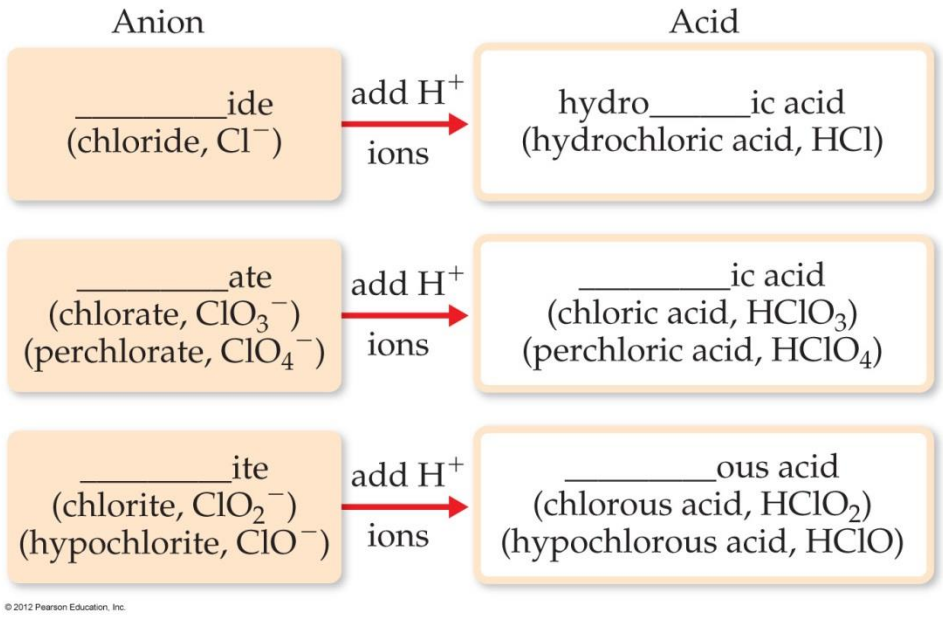


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- If the anion in the acid ends in ***-ite***, change the ending to ***-ous acid***.
  - HClO: hypochlorous acid
  - $HClO_2$: chlorous acid

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Acid Nomenclature



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- If the anion in the acid ends in ***-ate***, change the ending to ***-ic acid***.
  - $HClO_3$: chloric acid
  - $HClO_4$: perchloric acid

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Nomenclature of Binary Compounds

TABLE 2.6 • Prefixes Used in Naming Binary Compounds Formed between Nonmetals

Prefix	Meaning
Mono-	*1*
Di-	*2*
Tri-	*3*
Tetra-	*4*
Penta-	*5*
Hexa-	*6*
Hepta-	*7*
Octa-	*8*
Nona-	*9*
Deca-	*10*

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- The less electronegative atom is usually listed first.
- A prefix is used to denote the number of atoms of each element in the compound (***mono-*** is not used on the first element listed, however) .

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Nomenclature of Binary Compounds

TABLE 2.6 • Prefixes Used in Naming Binary Compounds Formed between Nonmetals

Prefix	Meaning
Mono-	*1*
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Tetra-	*4*
Penta-	*5*
Hexa-	*6*
Hepta-	*7*
Octa-	*8*
Nona-	*9*
Deca-	*10*

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- The ending on the more electronegative element is changed to ***-ide***.
  - CO_2: carbon dioxide
  - CCl_4: carbon tetrachloride

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Nomenclature of Binary Compounds

TABLE 2.6 • Prefixes Used in Naming Binary Compounds Formed between Nonmetals

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Mono-	*1*
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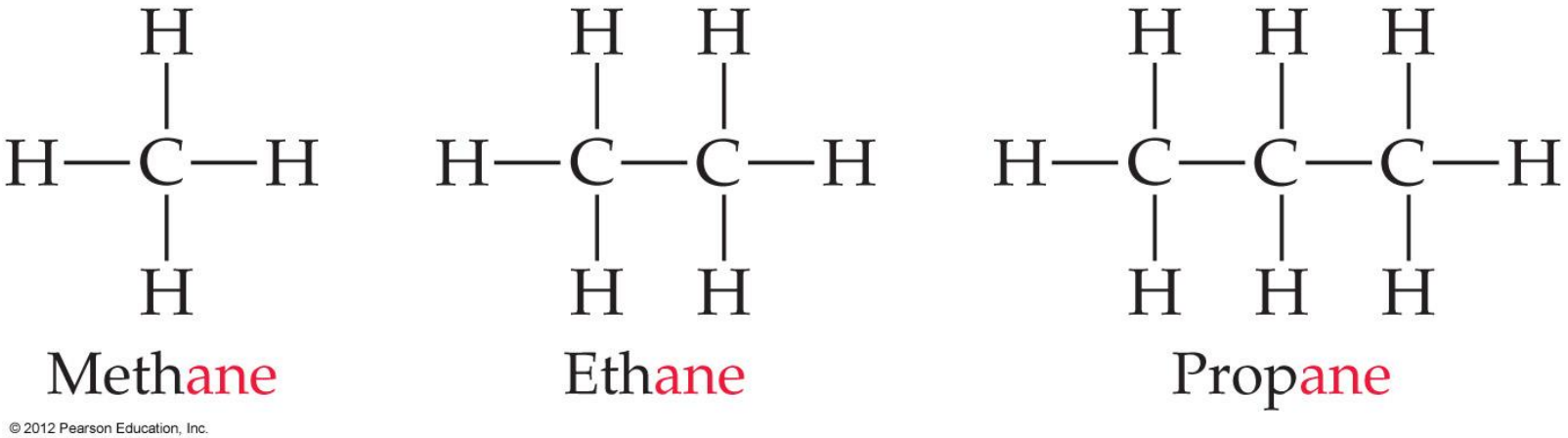
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- If the prefix ends with *a* or *o* and the name of the element begins with a vowel, the two successive vowels are often elided into one.

N_2O_5: dinitrogen pentoxide

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Nomenclature of Organic Compounds

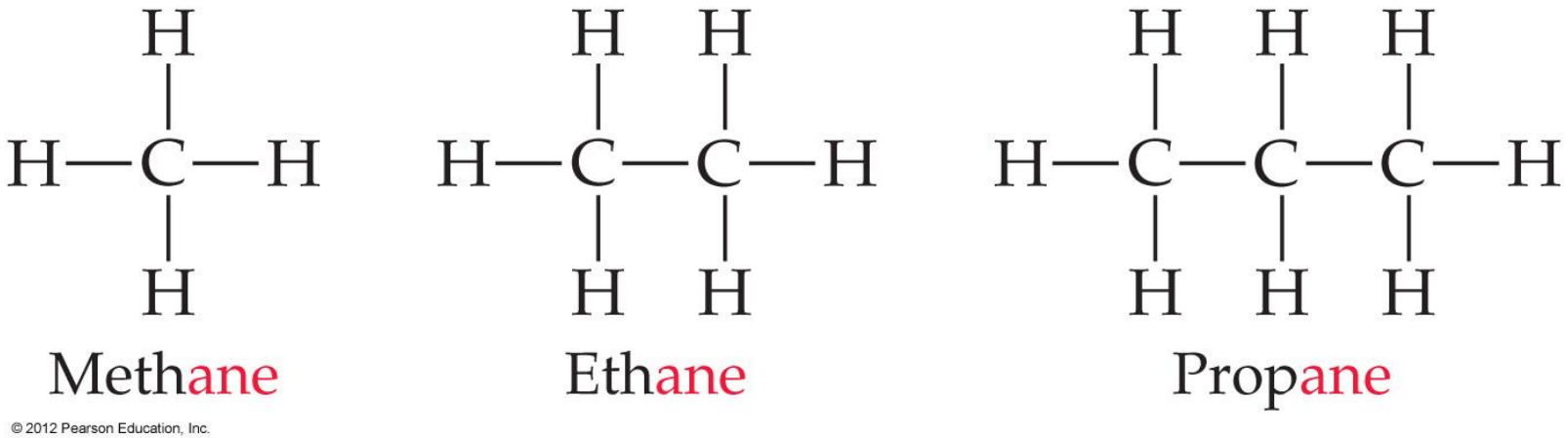


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- **Organic chemistry** is the study of carbon.
- Organic chemistry has its own system of nomenclature.

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Nomenclature of Organic Compounds

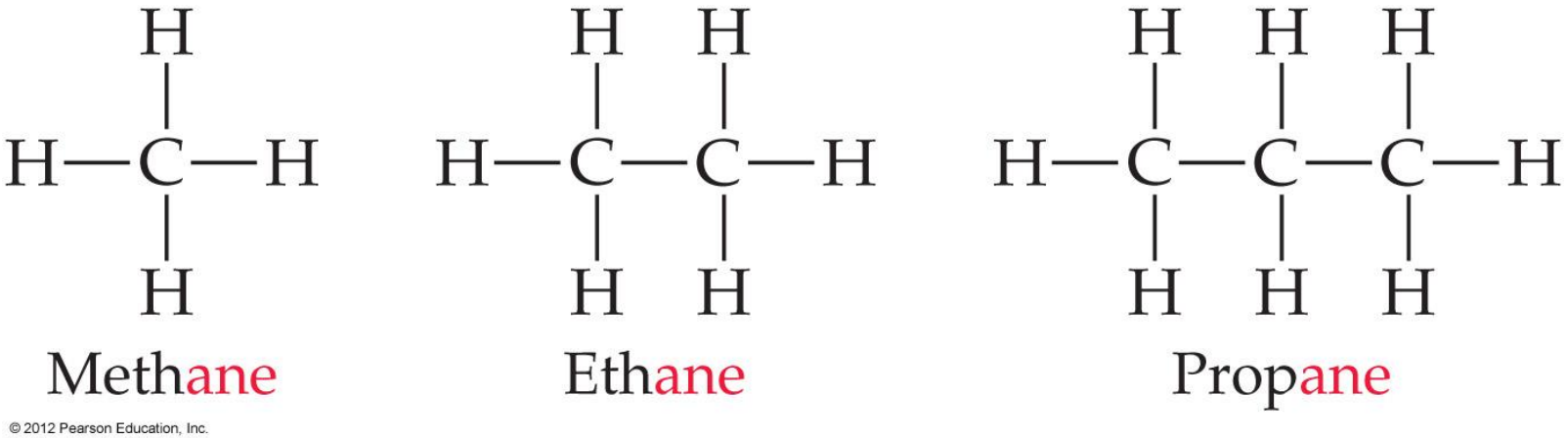


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The simplest hydrocarbons (compounds containing only carbon and hydrogen) are **alkanes**.

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Nomenclature of Organic Compounds

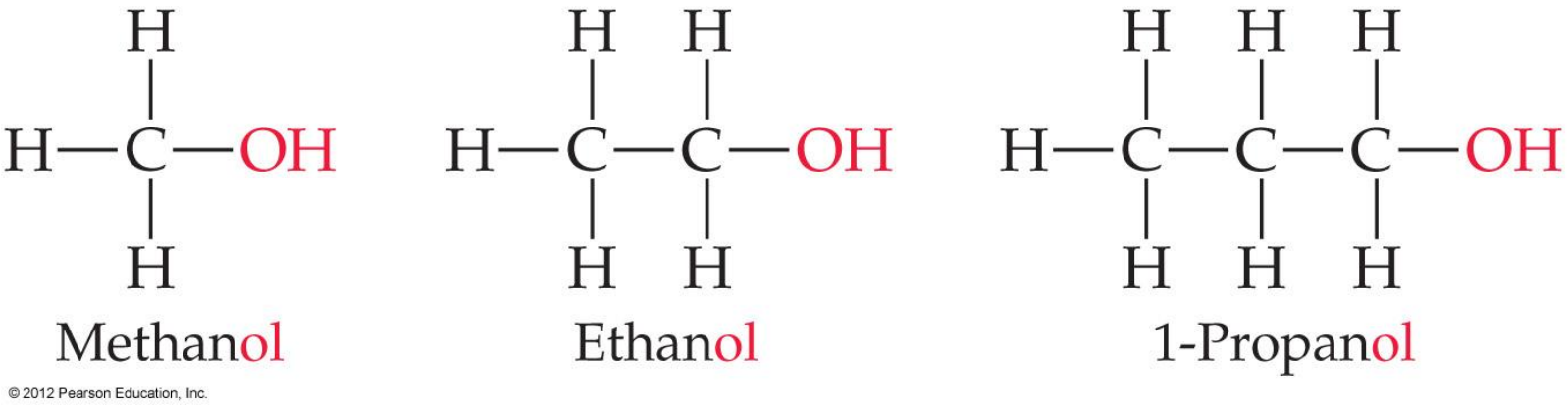


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The first part of the names just listed correspond to the number of carbons (***meth-*** = 1, ***eth-*** = 2, ***prop-*** = 3, etc.).

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Nomenclature of Organic Compounds



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- When a hydrogen in an alkane is replaced with something else (a **functional group**, like -OH in the compounds above), the name is derived from the name of the alkane.
- The ending denotes the type of compound.
  - An **alcohol** ends in *-ol*.

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